

4D Minimally Invasive Surgery Scene Reconstruction and Simulation Using Deformable 3D Gaussian Splatting

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ABSTRACT

Reconstructing deformable tissues from endoscopic videos is essential for surgical simulation, training, and automation. We present a 4D endoscopic reconstruction framework based on deformable 3D Gaussian Splatting (3DGS), featuring an integrated pipeline for 3DGS-to-closed-mesh conversion and tetrahedralization for solver-based physics simulation. We further demonstrate the practical utility of reconstructed tissues across multiple simulation engines, highlighting the pipeline's potential for interactive surgical education and high-fidelity, data-driven Extended Reality (XR) simulation systems.

Index Terms: 4D Reconstruction, Gaussian Splatting, Minimally Invasive Surgery, Medical Simulation

1 INTRODUCTION

Realistic reconstruction of robotic minimally invasive surgery (MIS) scenes is a foundational yet challenging problem. High-fidelity 4D models support downstream applications such as surgical simulation, robot autonomy, and XR-based training environments [8]. Traditional production pipelines rely on manually crafted soft-tissue assets, which are labor-intensive and limited in anatomical variability.

Data-driven reconstruction from real endoscopic videos offers a promising alternative but faces sparse viewpoints, constrained camera motion, heavy occlusion, rapid deformation, and topology changes. Recent neural reconstruction methods have advanced the state-of-the-art, yet significant room remains for improvement in achieving temporally coherent, photorealistic reconstruction with real-time rendering. Moreover, the lack of "out-of-the-box" tools and standardized automated pre-processing route for converting reconstructions (e.g., Gaussians) into simulation-ready closed tetrahedral meshes impedes physics-based modeling, limiting the efficacy of XR surgical education.

2 METHODOLOGY

This work builds upon a robust extension of the 3D Gaussian Splatting (3DGS) pipeline and emphasizes downstream medical simulation (see Figure 1). As illustrated in Figure 2, we adopt a deformable 3DGS framework optimized for surgical XR and soft-tissue simulation. By extending the real-time rasterization of 3DGS [2], our framework captures complex deformations and handles typical endoscopic challenges, such as tool occlusion.

2.1 Canonical Space Representation

A surgical scene is modeled in canonical space as a set of anisotropic 3D Gaussians $\{\mathcal{G}_c\}$. Each Gaussian is parameterized

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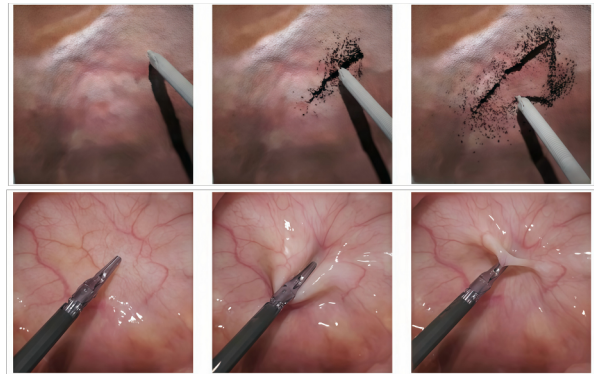


Figure 1: MPM and FEM simulation of reconstructed soft tissue for surgical training.

by center $\mu \in \mathbb{R}^3$, rotation $q \in \mathbb{R}^4$, scale $s \in \mathbb{R}^3$, opacity $\alpha \in \mathbb{R}$, and color $c \in \mathbb{R}^3$ (encoded via spherical harmonics).

To improve robustness in endoscopic settings (restricted viewpoints and persistent surgical tool occlusion), we employ a mask-guided, occlusion-aware initialization that enforces cross-frame re-projection consistency.

Final rendering is achieved via differentiable rasterization, where 3D Gaussians are projected onto the image plane to compute 2D covariance and aggregated through α -blending to produce photorealistic RGB and depth maps.

2.2 Observed Space Deformation

Instead of optimizing isolated static 3DGS instances at discrete timestamps and interpolating afterward, we introduce a unified deformable 3D Gaussian representation that models both geometry and motion through a continuous deformation field. A time-conditioned MLP predicts per-Gaussian deformation at any query time t_i , enabling temporally consistent rendering for XR playback.

2.3 Optimization

2.3.1 Occlusion-aware Appearance Modeling

Surgical instruments often cause prolonged occlusions, leading to incomplete color observations. Discrete Gaussians are sensitive to these visibility gaps. We apply targeted smoothing and adaptive normalization to Gaussians with low temporal visibility, stabilizing color prediction without explicit inpainting.

2.3.2 Temporal Flow Regularization

The reconstruction model Ψ is trained to render the corresponding image C_i at any timestamp t_i . Training jointly incorporates binary surgical-tool masks, stereo-derived depth maps, and optical-flow priors with edge preservation. These cues enhance geometric accuracy and stabilize non-rigid motion estimation for high-frequency tissue deformation.

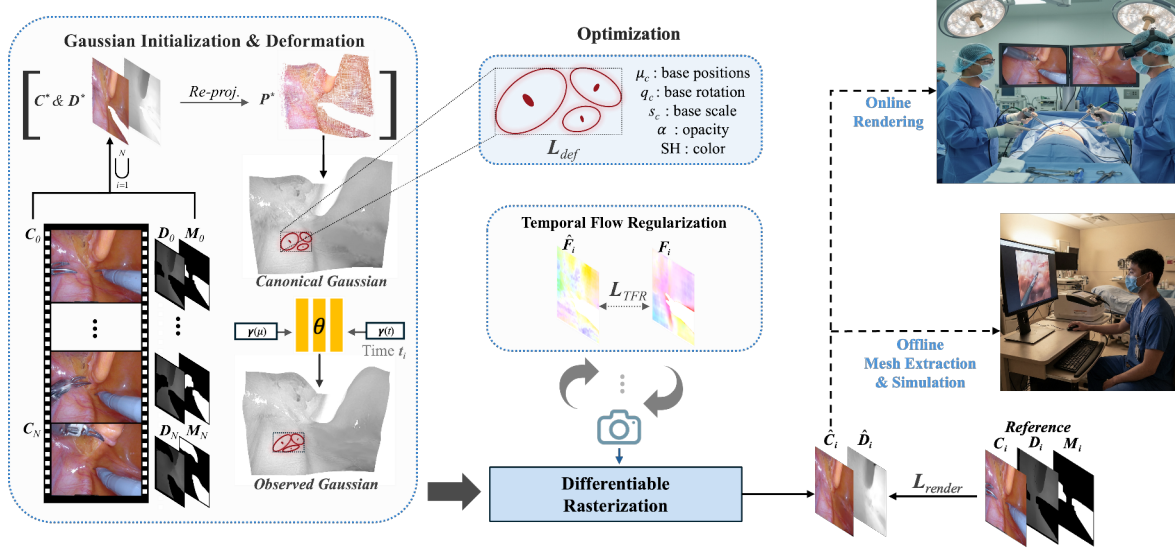


Figure 2: System overview of the proposed 4D reconstruction and simulation pipeline.

Table 1: Quantitative comparison on the EndoNeRF dataset [6] of our method against four prior SOTAs.

Models	EndoNeRF-Cutting			EndoNeRF-Pulling		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
EndoNeRF [6]	35.0442	0.9369	0.1462	36.1403	0.9474	0.1478
EndoGaussian [3]	27.1522	0.9302	0.1091	28.8755	0.9378	0.1050
Endo-4DGS [1]	36.5649	0.9552	0.0326	37.8538	0.9591	0.0435
HFGS [9]	24.1317	0.7909	0.0816	28.3212	0.8788	0.0646
Ours	38.0459	0.9656	0.0321	40.6364	0.9756	0.0266

2.4 Closed Mesh Extraction and Closure

To enable solver-based simulation, the sparse Gaussian cloud must be converted into a continuous manifold. Our pipeline applies bilateral filtering to suppress noise followed by masked Poisson surface reconstruction to generate an initial open mesh.

Since endoscopic views typically capture anterior surfaces, the resulting mesh is anatomically incomplete. We apply an efficient mesh-closure procedure that identifies boundary loops, extrudes a base surface along estimated vertex normals, and stitches the peripheral sides. This converts the “sheet-like” surface into a watertight manifold ready for volumetric discretization.

3 EXPERIMENTS AND RESULTS

We evaluate on two representative deformation cases (“pulling” and “cutting”) from the EndoNeRF dataset [6], using a 7:1 train-validation split. Reconstruction quality is assessed using PSNR, SSIM, and LPIPS metrics computed on tissue regions. As illustrated in Table 1, Our method achieves real-time rendering exceeding 70 FPS at 512×640 resolution on an RTX 3090 GPU and outperforms four recent methods [6] [3] [1] [9] across all major metrics.

We validate the integration of reconstructed geometry into XR surgical training pipelines through two simulations: a Taichi-based Material Point Method (MPM) solver using extracted tetrahedral solids [5], and a Finite Element Method (FEM) simulation based on our custom SOFA framework [4] [7]. These demos confirm the feasibility of using our reconstruction for interactive, data-driven simulation playback.

4 CONCLUSION

We present a deformable 3DGS-based framework for high-fidelity 4D surgical scene reconstruction, with an integrated pathway from Gaussians to watertight meshes and tetrahedral solids for solver-based simulation. This pipeline offers a practical solution for interactive surgical education and data-driven XR surgical training.

While demonstrating potential for XR pipelines, the current system is limited by offline calibration and preliminary fracture realism. To improve real2sim dynamics, future research will focus on integrating advanced simulation methods (XMPM/XFEM) and injecting physical priors into deformation field modeling. Overall, our pipeline establishes a key foundation for robot-assisted surgical applications and XR-oriented dynamic scene reconstruction.

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